

Predictive Maintenance of Industrial Machines Using Machine Learning

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DECLARATION

This Machine Learning project report is submitted to the Department of Computer Science, MNS-University of Agriculture, Multan, to partially fulfill the requirements of the semester project. We declare that this project work has been completed for academic purposes and has not been submitted elsewhere for any degree, diploma, or publication.

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ABSTRACT

Industrial machines rarely fail without warning signs. Sensor features such as air temperature, process temperature, rotational speed, torque, and tool wear can shift before breakdown. This project develops a supervised machine-learning workflow for early machine-failure prediction using the AI4I 2020 predictive-maintenance dataset structure. Five classifiers are trained and compared: Logistic Regression, K-Nearest Neighbors, Naive Bayes, Decision Tree, and Support Vector Machine. Evaluation uses accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and ROC curves. Recall and F1-score are

emphasized because a missed failure is usually more serious than a false maintenance alert.

Keywords: Predictive Maintenance, Industrial Machines, Machine Learning, KNN, Naive Bayes, Decision Tree, SVM, Logistic Regression.

LIST OF ABBREVIATIONS

MNSUAM: MNS-University of Agriculture, Multan

CS: Computer Science

ML: Machine Learning

KNN: K-Nearest Neighbors

SVM: Support Vector Machine

LR: Logistic Regression

DT: Decision Tree

AI4I: Artificial Intelligence for Industries

ROC-AUC: Receiver Operating Characteristic Area Under Curve

CHAPTER I: Introduction

1.1 Overview

Predictive maintenance is the use of data and machine-learning models to estimate failure risk before a machine breaks down. Traditional reactive maintenance starts after failure has already caused downtime. Preventive maintenance schedules service at fixed intervals, but this may replace parts too early or too late. Predictive maintenance is more flexible because it uses current machine-condition signals.

In industrial environments, sensors record temperature, speed, torque, and tool wear. These readings can contain early warning patterns. A machine-learning model can learn from historical examples and classify new machine states as normal or risky.

1.2 Project Goals

The aim of this project is to build and evaluate a predictive-maintenance model for industrial machines. The project compares multiple algorithms and selects the strongest model using balanced metrics. The main research question is: which machine-learning model can most reliably predict industrial machine failure using sensor features?

CHAPTER II: Literature Review

Predictive maintenance is an important application of Industry 4.0 and smart manufacturing. Data-driven maintenance systems use sensor readings to detect patterns that may not be visible through manual inspection. Classification algorithms are useful when the goal is to predict a binary target such as failure or no failure.

Logistic Regression is a common baseline for binary prediction because it estimates failure

probability. KNN compares new records with similar historical sensor records. Naive Bayes is a fast probabilistic classifier. Decision Tree models are easy to interpret because they produce rule-based splits. SVM can perform well when a clear separating boundary exists between classes.

Previous predictive-maintenance studies show that accuracy alone can be misleading when failures are rare. A model can achieve high accuracy by predicting the majority class while missing true failures. Therefore, recall, precision, F1-score, and ROC-AUC are important for model selection.

CHAPTER III: Project Description

3.1 Dataset Source

The intended dataset is the Kaggle AI4I 2020 predictive-maintenance dataset. It contains 10,000 machine records and 14 columns. The project includes a Kaggle API download script and an offline AI4I-compatible dataset generator for environments where Kaggle access is unavailable.

3.2 Dataset Features

The main input features are Type, Air temperature [K], Process temperature [K], Rotational speed [rpm], Torque [Nm], and Tool wear [min]. The target variable is Machine failure. The dataset also contains failure-mode columns: TWF, HDF, PWF, OSF, and RNF.

CHAPTER IV: Data Analysis and Processing

The preprocessing pipeline validates that all required columns exist. Identifier columns are kept in the dataset but excluded from model features. Machine type is categorical, so it is one-hot encoded. Numeric sensor columns are scaled for Logistic Regression, KNN, Naive Bayes, and SVM because these models are sensitive to feature magnitude. Decision Tree does not require scaling.

The data is split into training and testing sets using stratification so that failure and non-failure classes remain proportionally represented. Class imbalance is handled through class weights in Logistic Regression, Decision Tree, and SVM.

CHAPTER V: Algorithm and Performance Analysis

5.1 Logistic Regression

Logistic Regression estimates the probability that a machine will fail. It is used as a mathematical baseline and supports class weighting for imbalanced data.

5.2 K-Nearest Neighbors

KNN classifies a machine based on the labels of nearby records in feature space. It requires scaling because distance calculations are affected by feature magnitudes.

5.3 Naive Bayes

Naive Bayes applies Bayes' theorem and assumes features are conditionally independent given the class. It is fast and useful as a probabilistic baseline.

5.4 Decision Tree

Decision Tree learns interpretable rules such as high tool wear, high torque, or abnormal temperature relationships. It is easy to explain but can overfit if not constrained.

5.5 Support Vector Machine

SVM finds a separating boundary between failure and non-failure classes. The project uses an RBF kernel and probability estimates for ROC-AUC calculation.

5.6 Evaluation

The project evaluates models with accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and ROC curve. Recall is especially important because a false negative means a real machine failure was missed.

5.7 Results

Model results are generated by running python scripts/train_models.py.

CHAPTER VI: Conclusion

This project builds a complete predictive-maintenance workflow for industrial machines. It prepares an AI4I-style dataset, trains five machine-learning classifiers, compares them using balanced metrics, and generates report figures and a final academic report. The best model is selected using F1-score, recall, and ROC-AUC rather than accuracy alone. Future improvements may include Random Forest, XGBoost, SHAP feature explanation, real-time IoT integration, and dashboard-based maintenance alerts.

REFERENCES

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